Logic and Set Theory
Lecture N°2
1 Lic Data Science

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Logic
Quantifiers
The Universal Quantifier
The Existential Quantifier
Strategies for proofs
Direct proof
Proof by Contrapositive
Proof by Contradiction
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Summary

Logic

- Quantifiers
- The Universal Quantifier
- The Existential Quantifier
- Strategies for proofs

Mathematical texts often use the expressions “for all” and “there exists”, resp. In formal notation the following symbols are used:

Quantifier	Symbol
for all	$\forall$
there exists a	$\exists$
there exists exactly one	$\exists!$
there does not exists	$\nexists$

Definition

Let $P(x)$ be an open sentence on a set A. Then the proposition $(\forall x \in A)$ or $\forall x, P(x)$ can be read as "**For every** element x in A, $P(x)$ **holds**" or simply, "**For all** $x, P(x)$". The symbol $\forall$, which reads "**for all**" or "**for every**", is called **the universal quantifier**

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Example

The proposition for all natural number $n, n+2>5$ is false since

$$\{n : n+2>5\} = \{4,5,6,...\} \neq \mathbb{N}.$$

Definition

Let $P(x)$ be an open sentence on a set A. Then the proposition "$\exists x \in A) P(x)$" or "$\exists x, P(x)$" reads "**There exists** $x \in A$ such that $P(x)$ **holds**" or simply, "**For some** $x, P(x)$".

The symbol $\exists$, which reads "**there exists**" or "**for some**" or "**for at least**", is called **the existential quantifier.**

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The symbol $\exists$, which reads "**there exists**" or "**for some**" or "**for at least**", is called **the existential quantifier.**

Example

The proposition exists $n \in \mathbb{N}, n + 4 < 7$ is true since

$$\{n \in N : n + 4 < 7\} = \{1, 2\} \neq \emptyset.$$

Definition

For an open sentence $P(x)$, the proposition "$(\exists! x), P(x)$" is read "There exists a unique x, such that $P(x)$". The sentence "$(\exists! x), P(x)$" is true, when the truth set for $P(x)$ contains exactly one element from the universe; hence $\exists!$ is called the unique existence quantifier.

The negation of the proposition "All the integers are positive" reads "It is not true that all integers are positive"; in other words, there exists at least one integer, which is not positive.

Symbolically, if $\mathbb{Z}$ denotes the set of integer, then the above can be written as

$\neg(\forall x \in \mathbb{Z})(x \ \ is \ \ positive) \equiv (\exists x \in \mathbb{Z})(x \ \ is \ \ not \ \ positive)$.

Furthermore, if $P(x)$ denotes "x is positive", then the above can be written as

$\neg(\forall x \in \mathbb{Z})P(x) \equiv (\exists x \in \mathbb{Z})\neg P(x)$.

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Theorem (De Morgan)

If $P(x)$ is an open sentence, then

1. $\neg(\forall x \in \mathbb{Z})P(x) \equiv (\exists x \in \mathbb{Z})\neg P(x)$.
2. $\neg(\exists x \in \mathbb{Z})P(x) \equiv (\forall x \in \mathbb{Z})\neg P(x)$.

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Proof.

1. The sentence $\neg(\forall x)P(x)$ is true

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   iff $(\forall x)P(x)$ is false,
   iff the truth set of $P(x)$ is not the universe,

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Proof.

1. The sentence $\neg(\forall x)P(x)$ is true
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Proof.

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   iff $(\exists x)\neg P(x)$ is true.
   Thus, $\neg(\forall x)P(x)$ is true if and only if $(\exists x)\neg P(x)$ is true. So the proposition is true.

Proof.

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   Thus, $\neg(\forall x)P(x)$ is true if and only if $(\exists x)\neg P(x)$ is true. So the proposition is true.
2. Exercise.

□

Proposition

1. $(\forall x \in E, P(x) \wedge Q(x)) \Leftrightarrow$
   $((\forall x \in E / P(x)) \wedge (\forall x \in E, Q(x)))$.

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2. $(\forall x \in E, P(x) \vee Q(x)) \nRightarrow$
   $((\forall x \in E / P(x)) \vee (\forall x \in E, Q(x)))$. *But*
   $((\forall x \in E / P(x)) \vee (\forall x \in E, Q(x))) \Rightarrow (\forall x \in$
   $E, P(x) \vee Q(x))$.

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Proposition

1. $(\forall x \in E, P(x) \wedge Q(x)) \Leftrightarrow$
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2. $(\forall x \in E, P(x) \vee Q(x)) \not\Rightarrow$
   $((\forall x \in E / P(x)) \vee (\forall x \in E, Q(x)))$. *But*
   $((\forall x \in E / P(x)) \vee (\forall x \in E, Q(x))) \Rightarrow (\forall x \in$
   $E, P(x) \vee Q(x))$.
3. $(\exists x \in E, P(x) \wedge Q(x)) \Rightarrow$
   $((\exists x \in E, P(x)) \wedge (\exists x \in E, Q(x)))$. *But*
   $((\exists x \in E, P(x)) \wedge (\exists x \in E, Q(x))) \not\Rightarrow (\exists x \in$
   $E, P(x) \wedge Q(x))$.

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3. $(\exists x \in E, P(x) \wedge Q(x)) \Rightarrow$
   $((\exists x \in E, P(x)) \wedge (\exists x \in E, Q(x)))$. *But*
   $((\exists x \in E, P(x)) \wedge (\exists x \in E, Q(x))) \nRightarrow (\exists x \in$
   $E, P(x) \wedge Q(x))$.
4. $(\exists x \in E, P(x) \vee Q(x)) \Longleftrightarrow$
   $((\exists x \in E, P(x)) \vee (\exists x \in E, Q(x)))$.

Direct proof

The simplest form of proof for a statement of the form $P \Rightarrow Q$ is the **direct proof**. First assume that P is true. Produce a series of steps, each one following from the previous ones, that eventually leads to conclusion Q. It warrants the name "direct proof" only to distinguish it from other, more intricate, methods of proof.

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Direct proof

Assume P

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Direct proof

Assume P

$\vdots$

$\vdots$

Direct proof

Assume P

$\vdots$

$\vdots$

Therefore Q

Direct proof

Assume P

$\vdots$

$\vdots$

Therefore Q

Thus $P \Rightarrow Q$.

Direct proof

Assume P

$\vdots$

$\vdots$

Therefore Q

Thus $P \Rightarrow Q$.

Direct proof

Example

Prove that if x is even integer, then x^2 is even.

Proof.

Suppose x is an even integer

Direct proof

Example

Prove that if x is even integer, then x^2 is even.

Proof.

Suppose x is an even integer

Then $x = 2k$ for some integer k.

Direct proof

Example

Prove that if x is even integer, then x^2 is even.

Proof.

Suppose x is an even integer
Then $x = 2k$ for some integer k.
Thus $x^2 = 4k^2 = 2(2k^2) = 2t$ for some integer t.

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Direct proof

Example

Prove that if x is even integer, then x^2 is even.

Proof.

Suppose x is an even integer
Then $x = 2k$ for some integer k.
Thus $x^2 = 4k^2 = 2(2k^2) = 2t$ for some integer t.
Since x^2 is twice the integer t, x^2 is even.

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Direct proof

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Example

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Since x^2 is twice the integer t, x^2 is even. □

Direct proof

Example

prove that the sum of two even integers is always even : Consider two even integers x and y. Since they are even, they can be written as $x = 2a$ and $y = 2b$, respectively, for some integers a and b. Then the sum is $x + y = 2a + 2b = 2(a + b)$. Therefore $x + y$ has 2 as a factor and, by definition, is even. Hence, the sum of any two even integers is even. This proof uses the definition of even integers and the distributive property.

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Proof by Exhaustion

Proof by **exhaustion** (also known as **Proof by Cases**) is the proof that something is true by showing that it is true for each and every case that could possibly be considered. We can generalise the method of proof by exhaustion using two steps: 1. Establish the cases that apply to the statement. 2. Prove that the statement is true for each case. Each case of the statement needs to be proved separately, one by one. We need to exhaust all of the cases to verify the statement.

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Proof by Exhaustion

Example

Show that $\forall n \in \mathbb{N}, \frac{n(n+1}{2} \in \mathbb{N}$.

Proof.

- Step 1: Split the statement into a finite number of cases. Thus we can write $n = 2k$ or $n = 2k + 1$, with k being any natural number.
- Step 2: Now prove that the statement is true for each case.
  - case 1: n is even, $\exists k \in \mathbb{N}$ such as $n = 2k \Rightarrow \frac{2k(2k+1)}{2} = k(2k + 1) \in \mathbb{N}$.
  - case 2: n is odd, $\exists k \in \mathbb{N}$ such as $n = 2k + 1 \Rightarrow \frac{(2k+1)[(2k+1)+1]}{2} = \frac{(2k+1)(2k+2)}{2} = (2k + 1)(k + 1) \in \mathbb{N}$.



Exercise

Show that, if n is a positive integer, then $n(n^4 - 1)$ *is divisible by 2.*

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Proof by Contrapositive

A proof by **contrapositive** takes advantage of the mathematical equivalence $P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$. That is, a proof by contrapositive begins by assuming that Q is false (i.e., $\neg Q$ is true). It then produces a series of direct implications leading to the conclusion that P is false (i.e., $\neg P$ is true). It follows that Q cannot be false when P is true, so $P \Rightarrow Q$.

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Proof by Contrapositive:

Suppose $\neg Q$.

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Proof by Contrapositive:

Suppose $\neg Q$.

$\vdots$

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Proof by Contrapositive:

$$\textit{Suppose} \quad \neg Q.$$

$$\vdots$$

$$\textit{Therefore} \quad \neg P \quad (\textit{Via} \quad a \quad \textit{direct} \quad \textit{proof}).$$

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$$\textit{Suppose} \quad \neg Q.$$

$$\vdots$$

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$$\textit{Thus} \quad \neg Q \Rightarrow \neg P.$$

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Proof by Contrapositive:

$$Suppose \quad \neg Q.$$

$$\vdots$$

$$Therefore \quad \neg P \quad (Via \quad a \quad direct \quad proof).$$

$$Thus \quad \neg Q \Rightarrow \neg P.$$

$$Therefore \quad P \Rightarrow Q.$$

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Example

Prove that a^2 is even $\Rightarrow$ a is even.

Proof by Contrapositive:

Proof.

Consider the contrapositive

$$\neg(a \quad is \quad even) \Rightarrow \neg(a^2 \quad is \quad even).$$

Proof by Contrapositive:

Proof.

Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Proof by Contrapositive:

Proof.

Consider the contrapositive

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That is,

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Since a is odd, then

Proof by Contrapositive:

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Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Since a is odd, then$a = 2k + 1$, for some integer k.

Proof by Contrapositive:

Proof.

Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Since a is odd, then$a = 2k + 1$, for some integer k. Then

$$a^2 \quad = \qquad\qquad (2k+1)^2$$

Proof by Contrapositive:

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Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Since a is odd, then$a = 2k + 1$, for some integer k. Then

$$\begin{aligned} a^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \end{aligned}$$

Proof by Contrapositive:

Proof.

Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Since a is odd, then$a = 2k + 1$, for some integer k. Then

$$\begin{aligned} a^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

Proof by Contrapositive:

Proof.

Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Since a is odd, then$a = 2k + 1$, for some integer k. Then

$$\begin{aligned} a^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2r + 1, \quad where \ \ r \ \ is \ \ an \ \ integer. \end{aligned}$$

Proof by Contrapositive:

Proof.

Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Since a is odd, then$a = 2k + 1$, for some integer k. Then

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Thus a^2 is odd.

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Proof by Contrapositive:

Proof.

Consider the contrapositive

$$\neg(a \ \ is \ \ even) \Rightarrow \neg(a^2 \ \ is \ \ even).$$

That is,

$$a \ \ is \ \ odd \Rightarrow a^2 \ \ is \ \ odd.$$

Since a is odd, then$a = 2k + 1$, for some integer k. Then

$$\begin{aligned} a^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2r + 1, \quad where \ \ r \ \ is \ \ an \ \ integer. \end{aligned}$$

Thus a^2 is odd.

So, by contrapositive, if a^2 is even, then a is even.

□

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Proof by Contradiction:

A proof by **contradiction** is based on the mathematical equivalence $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$. In a proof by contradiction, one starts by assuming that both P and $\neg Q$ are true. Then, a series of direct implications are given that lead to a logical contradiction. Hence, $P \wedge \neg Q$ cannot be true and $P \Rightarrow Q$.

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Proof by Contradiction:

$$\textit{Suppose} \quad \neg P.$$

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Proof by Contradiction:

$$\textit{Suppose} \quad \neg P.$$

$$\vdots$$

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Proof by Contradiction:

Suppose $\neg P$.

$\vdots$

Therefore Q.

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Proof by Contradiction:

Suppose $\neg P$.

$\vdots$

Therefore Q.

$\vdots$

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Proof by Contradiction:

Suppose $\neg P$.

$\vdots$

Therefore Q.

$\vdots$

Therefore $\neg Q$.

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Proof by Contradiction:

$$\textit{Suppose} \quad \neg P.$$

$$\vdots$$

$$\textit{Therefore} \quad Q.$$

$$\vdots$$

$$\textit{Therefore} \quad \neg Q.$$

$$\textit{Hence} \quad Q \wedge \neg Q, \quad \textit{acontradiction}.$$

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Proof by Contradiction:

Suppose $\neg P$.

$\vdots$

Therefore Q.

$\vdots$

Therefore $\neg Q$.

Hence $Q \wedge \neg Q$, *acontradiction*.

Thus P.

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Example

Prove that $\sqrt{2}$ is an irrational number. Proof. We will prove this by contradiction. We assume $x = \sqrt{2}$ is a rational number. Then $x^2 = 2$. Since x is rational, there are integers n and m such that $x = \frac{n}{m}$, and n and m have no common factors other than 1 and -1. Since $x^2 = 2$, we get $n^2 = 2m^2$. Therefore, n^2 is even, hence n must be even. By definition of even integers, there exists an integer k such that $n = 2k$. Thus, we have $2k^2 = m^2$, and m must also be even. We thus have a contradiction, since m and n have 2 as common factor. Hence, $x = \sqrt{2}$ is not rational.

Proof By Counterexample

Recall $\neg\forall x P(x) \equiv \exists x, \neg P(x)$.
To establish that $\forall x P(x)$ is false (that is,$\neg\forall x P(x)$ is true) find a c such that $\neg P(c)$ is true (that is $P(c)$ is false). Such c is called a **counterexample** to the assertion $\neg\forall x P(x)$.

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Example

"Every positive integer is the sum of the squares of 3 integers."

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Example

"Every positive integer is the sum of the squares of 3 integers."
The integer 7 is a counterexample. So the claim is false

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Exercise (6: Worksheet N°1)

Let's consider two functions f and g from $\mathbb{R}$ to $\mathbb{R}$. We suppose that $f \mathring{u} g = 0$. Can we affirm that $f = 0$ or $g = 0$?

No, we cannot affirm that $f = 0$ or $g = 0$. The product of two functions $f \cdot g$ being zero does not mean that either function is identically zero. For example if

$$f(x) = \begin{cases} 1 - x^2 & \text{if} \quad x \in]0, +\infty[\\ 0 & \text{if} \quad x \in]-\infty, 0] \end{cases}$$

And

$$g(x) = \begin{cases} 0 & \text{if} \quad x \in]0, +\infty[\\ \frac{1}{1+x^2} & \text{if} \quad x \in]-\infty, 0] \end{cases}$$

Let's check the product $f \cdot g$: For $x \geq 0$:
$f(x)g(x) = 1 \cdot 0 = 0$. For $x < 0$: $f(x)g(x) = 0 \cdot 1 = 0$.

Exercise

Give a counterexample to disprove the statement :

All even numbers are not prime.

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Induction

The Principle of Mathematical Induction (PMI) Let P(n) be a statement about the positive integer n. If the following are true :

1. Base Case : We need to show that $p(n)$ is true for the smallest possible value of n, show that $p(n_0)$ is true.
2. Induction Hypothesis : Assume that the statement $p(n)$ is true for any positive integer n, for $n \geq n_0$.
3. Inductive Step : Show that the statement $p(n+1)$ is true.
4. Conclusion : We recall that by induction, $P(n)$ is true for all $n \in \mathbb{N}$.

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Proof by Induction

Example

Prove that for all positive integers n,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2},$$

i.e.,

$$P(n) = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Induction

Proof: Step 1: Let us check whether the formula is true for $n = 1$. Indeed

- $\sum_{i=1}^{1} i = 1.$

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$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \longrightarrow \sum_{i=1}^{n+1} i = \frac{(n+1)(n+2)}{2}.$$

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The sum for $n+1$ may be written as:

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The sum for $n+1$ may be written as:

$$\sum_{i=1}^{n+1} i \quad = \quad \sum_{i=1}^{n} i + (n+1)$$

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The sum for $n+1$ may be written as:

$$\begin{aligned} \sum_{i=1}^{n+1} i &= \sum_{i=1}^{n} i + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \end{aligned}$$

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$$\begin{aligned}\sum_{i=1}^{n+1} i &= \sum_{i=1}^{n} i + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1)+2(n+1)}{2}\end{aligned}$$

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\sum_{i=1}^{n+1} i &= \sum_{i=1}^{n} i + (n+1) \\
&= \frac{n(n+1)}{2} + (n+1) \\
&= \frac{n(n+1) + 2(n+1)}{2} \\
&= \frac{(n+1)(n+2)}{2}.
\end{aligned}$$

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\end{aligned}$$

So, we have shown that if $P(n)$ is true, then $P(n+1)$ is true

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Show that for any natural number $n \geq 4$, we have $2n < n!$.
Let us show by induction on the integer $n \geq 4$, that the property $P(n) : 2n < n!$ is true.

- Base case : For $n = 4$, we have $2^n = 16$ and $n! = 4! = 24$, which shows that $2^4 < 4!$. Thus, $P(n)$ is true for $n = 4$.
- Inductive Step : Assume that $P(n)$ is true and show that $P(n+1)$ is true, which amounts to assuming that $2^n < n!$ and to show that $2^{(n+1)} < (n+1)!$.
  We therefore assume that $2^n < n!$, then $2^{n+1} = 2.2^n < 2(n!)$, as $n \geq 4$, then $n > 2$ and also $n + 1 > 2$, which leads to $2^{n+1} < 2(n!) < (n+1).n! = (n+1)!$. So if $2^n < n!$ then $2^{n+1} < (n+1)!$, which shows that $P(n) \Rightarrow P(n+1)$.
- In conclusion, by induction on $n \geq 4$, we have $2^n < n!$.

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Exercise

Prove that, for all $n \in \mathbb{N}, 2^n > n$.

Exercise

Prove by induction that : $9^n - 1$ *is divisible by* 8 *for all non-negative integers.*

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Theorems that are Biconditional Statements

To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true.

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Explanation: :

$$(p \rightarrow q) \wedge (q \rightarrow p) \equiv p \leftrightarrow q$$

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Example

Prove that:
"If n is an integer, then n is odd if and only if n^2 is odd."

Proof:

We have already shown that both $p \Rightarrow q$ and $q \Rightarrow p$ are true.

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Proof:

We have already shown that both $p \Rightarrow q$ and $q \Rightarrow p$ are true.

$p \Rightarrow q$) Suppose n is an even integer

Then $n = 2k$ for some integer k.

Thus $n^2 = 4k^2 = 2(2k^2) = 2t$ for some integer t. Since n^2 is twice the integer t, n^2 is even.

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$q \Rightarrow p$) Consider the contrapositive

$$\neg(n \ \ is \ \ even) \Rightarrow \neg(n^2 \ \ is \ \ even).$$

That is,

$$n \ \ is \ \ odd \Rightarrow n^2 \ \ is \ \ odd.$$

Since n is odd, then $n = 2k + 1$, for some integer k. Then

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2r + 1, \quad where \ \ r \ \ is \ \ an \ \ integer. \end{aligned}$$

Thus n^2 is odd.

So, by contrapositive, if n^2 is even, then n is even..

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Therefore, $p \Leftrightarrow q$.